
Valuations on fields



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Contents

1	Valuation fields	1
1.1	Absolute values and completion	1
1.2	Places on a field	5
2	Non-archimedean valuations	7
2.1	Topology: Ultra-metric space	7
2.2	Algebra: ring of integers and residue field	9
2.3	Hensel's Lemma	11
2.4	Newton polygons	12
3	Finite field extensions	12
3.1	Finite-dimensional vector space	12
3.2	Finite field extensions	15
3.3	Ramification and inertia	16
4	Example: p-adic fields	17
4.1	p -adic fields	17
4.2	Completion	18
	References	19

1 Valuation fields

1.1 Absolute values and completion

Definition 1.1. Let $\mathbf{k}$ be a field. An *absolute value* on $\mathbf{k}$ is a function $|\cdot| : \mathbf{k} \rightarrow \mathbb{R}_{\geq 0}$ satisfying the following properties for all $x, y \in \mathbf{k}$:

- (a) $|x| = 0$ if and only if $x = 0$;
- (b) $|xy| = |x| \cdot |y|$;
- (c) $|x + y| \leq |x| + |y|$.

A field $\mathbf{k}$ equipped with an absolute value $|\cdot|$ is called a *valuation field*.

Remark 1.2. Let $\mathbf{k}$ be a field. Recall that a (additive) valuation on $\mathbf{k}$ is a function $v : \mathbf{k}^\times \rightarrow \mathbb{R}$ such that

- $\forall x, y \in \mathbf{k}^\times, v(xy) = v(x) + v(y)$;
- $\forall x, y \in \mathbf{k}^\times, v(x + y) \geq \min\{v(x), v(y)\}$.

We can extend v to the whole field $\mathbf{k}$ by defining $v(0) = +\infty$. Fix a real number $\varepsilon \in (0, 1)$. Then v induces an absolute value $|\cdot|_v : \mathbf{k} \rightarrow \mathbb{R}_+$ defined by $|x|_v = \varepsilon^{v(x)}$ for each $x \in \mathbf{k}$.

The valuation v defined above is called an *additive valuation*. And an absolute value $|\cdot|$ on $\mathbf{k}$ is called a *multiplicative valuation*. In this note, the term *valuation* may refer to either an additive valuation or a multiplicative valuation, depending on the context.

Example 1.3. Let $\mathbf{k}$ be a field. The *trivial absolute value* on $\mathbf{k}$ is defined as

$$|x| := \begin{cases} 0, & x = 0; \\ 1, & x \neq 0. \end{cases}$$

Example 1.4. Let $|\cdot|_\infty$ be the usual absolute value on the field $\mathbb{Q}$ of rational numbers. Then $(\mathbb{Q}, |\cdot|_\infty)$ is a valuation field. Its completion is the field $\mathbb{R}$ of real numbers equipped with the usual absolute value.

Example 1.5. Let p be a prime number. The valuation v_p on $\mathbb{Q}$ induced by the ideal $(p) \subset \mathbb{Z}$ is called the *p-adic valuation*. It gives an absolute value $|\cdot|_p$ on $\mathbb{Q}$; see [Example 4.2](#)

Definition 1.6. The *(multiplicative) valuation group* of a valuation field $(\mathbf{k}, |\cdot|)$ is defined as the subgroup of $\mathbb{R}_{>0}$ given by

$$|\mathbf{k}^\times| := \{|x| : x \in \mathbf{k}^\times\}.$$

We use the notation $\sqrt{|\mathbf{k}^\times|}$ to denote the set $\{|x|^{1/n} : x \in \mathbf{k}^\times, n \in \mathbb{Z}_{>0}\}$.

Note that an absolute value $|\cdot|$ is non-trivial if and only if its valuation group $|\mathbf{k}^\times|$ is not equal to $\{1\}$.

Definition 1.7. Let $(\mathbf{k}, |\cdot|)$ be a valuation field. We say that $\mathbf{k}$ is *complete* if the metric $d(x, y) := |x - y|$ makes $\mathbf{k}$ a complete metric space.

Lemma 1.8. Let $(\mathbf{k}, |\cdot|)$ be a valuation field and $(\widehat{\mathbf{k}}, |\cdot|)$ its completion as a metric space. Then the operations of addition and multiplication on $\mathbf{k}$ can be extended to $\widehat{\mathbf{k}}$ uniquely, making $(\widehat{\mathbf{k}}, |\cdot|)$ a complete valuation field containing $\mathbf{k}$ as a dense subfield.

Proof. Note that the operations of addition and multiplication on $\mathbf{k}$ are uniformly continuous with respect to the metric $d(x, y) = |x - y|$. Thus they can be extended to $\widehat{\mathbf{k}}$ uniquely. $\square$

Proposition 1.9. Let $(\mathbf{k}, |\cdot|)$ be a complete valuation field with non-trivial absolute value. Then $\mathbf{k}$ is uncountable.

Proof. Since the absolute value $|\cdot|$ is non-trivial, we can construct a sequence $\{x_n\}_{n=1}^\infty \subseteq \mathbf{k}$ inductively such that $|x_n| < |x_{n-1}|/2$ for any $n \geq 1$ and $|x_0| < 1$. Then there is an injective map from $\mathbb{N}^{\{0,1\}}$ to $\mathbf{k}$ defined by

$$(a_n)_{n=1}^\infty \mapsto \sum_{n=1}^\infty a_n x_n, \quad a_n \in \{0, 1\}.$$

Since $|x_n| < 2^{-n}$, the series $\sum_{n=1}^\infty a_n x_n$ converges in $\mathbf{k}$. Note $|x_n| > |\sum_{m \geq n} x_m|$ for each n , we have that the map is injective. Thus $\mathbf{k}$ is uncountable. $\square$

Unlike the real number field $\mathbb{R}$, even a valuation field is complete, we can not expect the theorem of

nested intervals to hold.

Definition 1.10. A valuation field $(\mathbf{k}, |\cdot|)$ is called *spherically complete* if every decreasing sequence of closed balls in $\mathbf{k}$ has a non-empty intersection.

Example 1.11. The field $\mathbb{C}_p$ of p -adic complex numbers is not spherically complete; see Proposition 4.6.

Definition 1.12. Let $(\mathbf{k}, |\cdot|)$ be a valuation field. We say that $\mathbf{k}$ is *non-archimedean* if its absolute value $|\cdot|$ satisfies the *strong triangle inequality*:

$$|x + y| \leq \max\{|x|, |y|\}, \quad \forall x, y \in \mathbf{k}.$$

Otherwise, we say that $\mathbf{k}$ is *archimedean*.

Example 1.13. Let v be an additive valuation on a field $\mathbf{k}$. Then the induced absolute value $|\cdot|_v$ as in Remark 1.2 is non-archimedean.

The converse is also true: if $(\mathbf{k}, |\cdot|)$ is a non-archimedean valuation field, then the function $v : \mathbf{k}^\times \rightarrow \mathbb{R}$ defined by $v(x) = -\log|x|$ is an additive valuation on $\mathbf{k}$.

The following proposition explains the terminology "archimedean".

Proposition 1.14. Let $(\mathbf{k}, |\cdot|)$ be a valuation field. Then $\mathbf{k}$ is archimedean if and only if the set $\{|n \cdot 1| : n \in \mathbb{Z}\}$ is unbounded.

Proof. If $\mathbf{k}$ is non-archimedean, then for any integer $n > 0$, we have

$$|n \cdot 1| = |1 + 1 + \cdots + 1| \leq \max\{|1|, |1|, \dots, |1|\} = 1.$$

Thus the set $\{|n \cdot 1| : n \in \mathbb{Z}\}$ is bounded.

Conversely, suppose that $\{|n \cdot 1| : n \in \mathbb{Z}\}$ is bounded by a real number $C > 0$. For any $x, y \in \mathbf{k}$, we have

$$|x + y|^n = \left| \sum_{k=0}^n \binom{n}{k} x^k y^{n-k} \right| \leq \sum_{k=0}^n C |x|^k |y|^{n-k}.$$

Without loss of generality, we assume that $|x| \leq 1$ and $y = 1$. Then the above inequality becomes

$$|x + 1|^n \leq \sum_{k=0}^n C |x|^k \leq C(n + 1).$$

Taking n -th root and letting $n \rightarrow \infty$, we obtain $|x + 1| \leq 1$. Hence $|x + y| \leq \max\{|x|, |y|\}$ for any $x, y \in \mathbf{k}$. $\square$

Theorem 1.15 (Ostrowski). Let $(\mathbf{k}, |\cdot|)$ be an archimedean complete valuation field. Then there exists an embedding $\iota : \mathbf{k} \hookrightarrow \mathbb{C}$ and $s \in (0, \infty)$ such that

$$|x| = |\iota(x)|_\infty^s, \quad \forall x \in \mathbf{k},$$

where $|\cdot|_\infty$ is the usual absolute value on $\mathbb{C}$. In other words, complete archimedean valuation fields are equivalent to either $\mathbb{R}$ or $\mathbb{C}$ equipped with usual absolute value (see Definition 1.16).

Proof. By Proposition 1.14, $\mathbf{k}$ has characteristic zero. For any integers $n, m \in \mathbb{Z}_{>0}$, we write n in base m as

$$n = a_k m^k + a_{k-1} m^{k-1} + \cdots + a_1 \cdot m + a_0,$$

where $a_i \in \{0, 1, \dots, m-1\}$ for each i and $a_k \neq 0$. Then we have

$$|n| \leq \sum_i |a_i| |m|^i \leq k |m|^{k+1} \quad \text{and} \quad k \leq \frac{\log n}{\log m}.$$

Combining the above two inequalities, we obtain

$$\log |n| \leq \log(\log n) + \left(\frac{\log n}{\log m} + 1 \right) \log |m|.$$

Replace n by n^t , $t \in \mathbb{Z}_{>0}$, and divide both sides by t , we get

$$\log |n| \leq \frac{\log(\log n) + \log t}{t} + \left(\frac{\log n}{\log m} + \frac{1}{t} \right) \log |m|.$$

Let t trend to infinity, we get

$$\frac{\log |n|}{\log n} \leq \frac{\log |m|}{\log m}.$$

By symmetry, we have

$$\frac{\log |n|}{\log n} = \frac{\log |m|}{\log m}, \quad \forall n, m \in \mathbb{Z}_{>0}.$$

Set s be the constant $\log |n| / \log n$ for any $n > 1$. Then we have $|n| = n^s$ for any $n \in \mathbb{Z}$. By Proposition 1.14, we get $s > 0$. Without loss of generality, we may assume that $s = 1$ by replacing $|\cdot|$ by $|\cdot|^{1/s}$.

We have show that the natural embedding $\mathbb{Q} \hookrightarrow \mathbf{k}$ is an isometry if we equip $\mathbb{Q}$ with the usual absolute value $|\cdot|_\infty$. Thus it can be extended to an isometric embedding $\mathbb{R} \hookrightarrow \mathbf{k}$. For every $\alpha \in \mathbf{k}$, consider

$$f(z) := |\alpha^2 - (z + \bar{z})\alpha + z\bar{z}|, \quad z \in \mathbb{C}.$$

Since $\lim_{z \rightarrow \infty} f(z) = +\infty$, there exists $z_0 \in \mathbb{C}$ such that $f(z_0) = m$ is the minimum of f . Since $f^{-1}(m)$ is closed and bounded, this is a compact set; and then we can assume that z_0 has the largest norm among all points in $f^{-1}(m)$. We only need to show that $m = 0$.

Suppose the contrary that $m > 0$. Choose $0 < \varepsilon < m$. The polynomial $g(T) = T^2 - (z_0 + \bar{z}_0)T + z_0\bar{z}_0 + \varepsilon$ has two distinct roots $\beta, \bar{\beta} \in \mathbb{C}$. Then $\beta\bar{\beta} = z_0\bar{z}_0 + \varepsilon$ implies $|\beta| > |z_0|$; and hence $f(\beta) > m$.

Fix $n \in \mathbb{Z}_{>0}$, define

$$G(T) = (T^2 - (z_0 + \bar{z}_0)T + z_0\bar{z}_0)^n - (-\varepsilon)^n.$$

We have $G(\beta) = 0$ and $G(T) = g(T) \prod_{i=1}^{n-1} (T^2 - (\beta_i + \bar{\beta}_i)T + \beta_i\bar{\beta}_i)$ for some $\beta_i \in \mathbb{C}$. Then

$$f(\beta)m^{n-1} \leq |G(\alpha)| \leq |\alpha^2 - (z_0 + \bar{z}_0)\alpha + z_0\bar{z}_0|^n + \varepsilon^n = m^n + \varepsilon^n.$$

It follows that

$$f(\beta) \leq m + \frac{\varepsilon^n}{m^{n-1}}.$$

Let n trend to infinity, we get $f(\beta) \leq m$, a contradiction. □

1.2 Places on a field

Definition 1.16. Let $\mathbf{k}$ be a field. Two absolute values $|\cdot|_1$ and $|\cdot|_2$ on $\mathbf{k}$ are said to be *equivalent* if there exists a real number $c \in (0, \infty)$ such that

$$|x|_1 = |x|_2^c, \quad \forall x \in \mathbf{k}.$$

Note that equivalent absolute values induce the same topology on the field $\mathbf{k}$. Moreover, the following lemma shows that the converse is also true.

Lemma 1.17. Let $\mathbf{k}$ be a field and $|\cdot|_1, |\cdot|_2$ be two absolute values on $\mathbf{k}$. Then the following statements are equivalent:

- (a) $|\cdot|_1$ and $|\cdot|_2$ are equivalent;
- (b) $|\cdot|_1$ and $|\cdot|_2$ induce the same topology on $\mathbf{k}$;
- (c) The unit disks $D_1 = \{x \in \mathbf{k} : |x|_1 < 1\}$ and $D_2 = \{x \in \mathbf{k} : |x|_2 < 1\}$ are the same.

Proof. The implications (a) $\Rightarrow$ (b) is obvious. Now we prove (b) $\Rightarrow$ (c). For any $x \in D_1$, we have $x^n \rightarrow 0$ as $n \rightarrow \infty$ under the absolute value $|\cdot|_1$ and thus under $|\cdot|_2$. Therefore, $|x|_2^n \rightarrow 0$ as $n \rightarrow \infty$, which implies that $|x|_2 < 1$, i.e., $x \in D_2$. Similarly, we can prove that $D_2 \subseteq D_1$.

Finally, we prove (c) $\Rightarrow$ (a). If $|\cdot|_1$ is trivial, then $D_1 = \{0\}$ and thus $|\cdot|_2$ is also trivial. In this case, they are equivalent. Suppose that both $|\cdot|_1$ and $|\cdot|_2$ are non-trivial. Pick any $x, y \notin D_1 = D_2$. Then there exist real numbers $\alpha, \beta > 0$ such that $|x|_1 = |x|_2^\alpha$ and $|y|_1 = |y|_2^\beta$. Suppose the contrary that $\alpha \neq \beta$. Consider the domain $\Lambda \subseteq \mathbb{Z}^2$ defined by

$$\begin{cases} n \log |x|_2 < m \log |y|_2; \\ n\alpha \log |x|_2 > m\beta \log |y|_2. \end{cases}$$

Since $\alpha \neq \beta$, the two lines defined by the equalities are not parallel. Thus Λ is non-empty. Pick $(n, m) \in \Lambda$ and set $z := x^n y^{-m}$. Then we have $|z|_2 < 1$ and $|z|_1 > 1$, a contradiction. $\square$

Definition 1.18. Let $\mathbf{k}$ be a field. A *place* on $\mathbf{k}$ is an equivalence class of non-trivial absolute values on $\mathbf{k}$. An archimedean (resp. non-archimedean) place is also called an *finite* (resp. *infinite*) place. We denote the set of all places (resp. finite, infinite places) on $\mathbf{k}$ by $M_{\mathbf{k}}$ (resp. $M_{\mathbf{k}}^f, M_{\mathbf{k}}^\infty$). If $\mathbf{l} \subset \mathbf{k}$ is a subfield, we denote by $M_{\mathbf{k}/\mathbf{l}}$ the set of all places on $\mathbf{k}$ which are trivial on $\mathbf{l}$.

Example 1.19. Let $\mathbf{k} = \mathbb{C}(t)$ and $\mathbf{l} = \mathbb{C}$. Then I claim that

$$M_{\mathbb{C}(t)/\mathbb{C}} \cong \{ \text{prime divisors on } \mathbb{P}_{\mathbb{C}}^1 \}.$$

For each prime divisor P on $\mathbb{P}_{\mathbb{C}}^1$, we can define an additive valuation $\text{Mult}_P : \mathbb{C}(t)^\times \rightarrow \mathbb{Z}$ by assigning to each non-zero rational function $f \in \mathbb{C}(t)^\times$ its multiplicity at P . Fix a real number $\varepsilon \in (0, 1)$. Then we obtain a multiplicative valuation (absolute value) $|\cdot|_P$ on $\mathbb{C}(t)$ as in [Remark 1.2](#). It is easy to check that the absolute value $|\cdot|_P$ is trivial on $\mathbb{C}$ and that different prime divisors give rise to inequivalent absolute values.

Conversely, given any non-trivial absolute value $|\cdot|$ on $\mathbb{C}(t)$ which is trivial on $\mathbb{C}$, by [Proposi-](#)

tion 1.14 and [Example 1.13](#), the absolute value $|\cdot|$ is given by an additive valuation $v : \mathbb{C}(t)^\times \rightarrow \mathbb{R}$. Let $\mathcal{O}_v$ be the valuation ring of v and $\mathfrak{m}_v$ its maximal ideal. Then $t \in \mathcal{O}_v$ or $t^{-1} \in \mathcal{O}_v$. Without loss of generality, we assume that $t \in \mathcal{O}_v$. Since $|\cdot|$ is trivial on $\mathbb{C}$, we have $\mathbb{C}[t] \subseteq \mathcal{O}_v$. And we have $\mathfrak{m}_v \cap \mathbb{C}[t] \neq 0$ since otherwise v is trivial on $\mathbb{C}[t]^\times$ and thus on $\mathbb{C}(t)^\times$. It follows that the image of $\mathbb{C}[t]$ under the quotient map $\mathcal{O}_v \rightarrow \mathcal{O}_v/\mathfrak{m}_v$ is $\mathbb{C}$. This gives a point $P \in \mathbb{A}_{\mathbb{C}}^1 \subseteq \mathbb{P}_{\mathbb{C}}^1$. Then v is different from the additive valuation Mult_P by a positive scalar multiple via looking at the values on $\mathbb{C}[t]$. Thus we have established the claimed bijection.

Theorem 1.20 (Ostrowski). Every nontrivial absolute value on $\mathbb{Q}$ is equivalent to either the usual absolute value $|\cdot|_\infty$ or a p -adic absolute value $|\cdot|_p$ for some prime number p . In other words,

$$M_{\mathbb{Q}} = \{|\cdot|_\infty\} \cup \{|\cdot|_p : p \text{ is a prime number}\}.$$

Proof. By [Theorem 1.15](#), all archimedean places on $\mathbb{Q}$ are equivalent to the usual absolute value $|\cdot|_\infty$. Now we consider a non-archimedean absolute value $|\cdot|$ on $\mathbb{Q}$. By [Example 1.13](#), $|\cdot|$ is induced by an additive valuation $v : \mathbb{Q}^\times \rightarrow \mathbb{R}$. Consider its valuation ring $\mathcal{O}_v$ and maximal ideal $\mathfrak{m}_v$. We have $\mathbb{Z} \subseteq \mathcal{O}_v$. If $\mathfrak{m}_v \cap \mathbb{Z} = 0$, then $\mathbb{Q} \subset \mathcal{O}_v$ and thus v is trivial, a contradiction. Thus there exists a prime number p such that $\mathfrak{m}_v \cap \mathbb{Z} = p\mathbb{Z}$. It follows that v is different from the p -adic valuation v_p by a positive scalar multiple via looking at the values on $\mathbb{Z}$. Hence $|\cdot|$ is equivalent to the p -adic absolute value $|\cdot|_p$. $\square$

Remark 1.21. For every non-archimedean place v on $\mathbb{Q}$ corresponding to a prime number p , we choose the unique normalized absolute value $|\cdot|_v$ in the class v such that $|p|_v = 1/p$. For the archimedean place v on $\mathbb{Q}$, we choose the usual absolute value $|\cdot|_v = |\cdot|_\infty$. Unless otherwise specified, we always use the normalized absolute values on $\mathbb{Q}$.

Remark 1.22. For any non-zero rational number $x \in \mathbb{Q}^\times$, one can easily check the *product formula*:

$$\prod_{v \in M_{\mathbb{Q}}} |x|_v = 1.$$

This can be viewed as an arithmetic analogue of the fact on $\mathbb{P}_{\mathbb{C}}^1$ that

$$\sum_{P \in \mathbb{P}^1(\mathbb{C})} \text{Mult}_P(f) = 0$$

for any non-zero rational function $f \in \mathbb{C}(t)^\times$. Indeed, fix a real number $\varepsilon \in (0, 1)$. Then by [Example 1.19](#), above fact can be rewritten as

$$\prod_{P \in \mathbb{P}_{\mathbb{C}}^1} |f|_P = 1.$$

Theorem 1.23 (Artin-Whaples approximations). Let $\mathbf{k}$ be a field and $v_1, v_2, \dots, v_n \in M_{\mathbf{k}}$ be pairwise distinct places on $\mathbf{k}$. For each $i \in \{1, 2, \dots, n\}$, let $x_i \in \mathbf{k}$ and $\varepsilon_i > 0$. Then there exists an element $x \in \mathbf{k}$ such that

$$|x - x_i|_{v_i} < \varepsilon_i, \quad \forall i \in \{1, 2, \dots, n\}.$$

In particular, the image of the diagonal embedding

$$\mathbf{k} \rightarrow \prod_{i=1}^n \mathbf{k}_{v_i}$$

is dense, where $\mathbf{k}_{v_i}$ is the completion of $\mathbf{k}$ with respect to the place v_i , and the product is equipped with the product topology.

Proof. We only need to find $c_i \in \mathbf{k}$ such that

$$|c_i|_{v_j} < 1, \quad \forall j \neq i; \quad \text{and} \quad |c_i|_{v_i} > 1.$$

If so, set $c_i(m) = c_i^m / (1 + c_i^m)$ for each $m \in \mathbb{Z}_{>0}$. Then we have

$$\lim_{m \rightarrow \infty} c_i(m) = \begin{cases} 0, & \text{under } v_j, j \neq i; \\ 1, & \text{under } v_i. \end{cases}$$

Thus the element $x = \sum_{i=1}^n c_i(m)x_i$ satisfies the required property for sufficiently large m .

Now we construct such $c = c_1$ by induction on n . By proof of [Lemma 1.17](#), there exists c with $|c|_1 > 1$ and $|c|_2 < 1$. This gives the base case $n = 2$. Suppose that we have constructed c for $n - 1$ places, i.e., $|c|_1 > 1$ and $|c|_i < 1$ for all $2 \leq i \leq n - 1$. There is $a \in \mathbf{k}$ such that $|a|_1 > 1$ and $|a|_n < 1$. Consider

$$c_m = \begin{cases} \frac{c^m}{c^{m+1}}, & |c|_n > 1; \\ c^m, & |c|_n = 1; \end{cases}$$

we have

$$\lim_{m \rightarrow \infty} |ac_m|_i = \begin{cases} +\infty, & i = 1; \\ 0, & i = 2, \dots, n - 1; \\ |a|_n < 1, & i = n. \end{cases}$$

Thus for sufficiently large m , the element $c = ac_m$ satisfies the required property. $\square$

2 Non-archimedean valuations

2.1 Topology: Ultra-metric space

We will use $B(x, r)$ (resp. $E(x, r)$) to denote the open ball (resp. closed ball) with center x and radius r .

Definition 2.1. A metric space (X, d) is called an *ultra-metric space* if its metric d satisfies the *strong triangle inequality*:

$$d(x, z) \leq \max\{d(x, y), d(y, z)\}, \quad \forall x, y, z \in X.$$

Remark 2.2. The term *ultra-metric space* should be translated into Chinese as “奥特度量空间”. There is no special reason for this translation, except that I insist on using “奥特” to translate “ultra”.

If $(\mathbf{k}, \|\cdot\|)$ is a non-archimedean field, then the metric $d(x, y) := \|x - y\|$ on $\mathbf{k}$ makes $(\mathbf{k}, d)$ an ultra-metric space.

Proposition 2.3. Let (X, d) be an ultra-metric space. Then for any $x, y, z \in X$, at least two of the three distances $d(x, y), d(y, z), d(z, x)$ are equal. And the third distance is less than or equal to the common value of the other two.

Proof. Suppose that $d(x, y) \geq d(y, z)$. By the strong triangle inequality, we have

$$d(x, z) \leq \max\{d(x, y), d(y, z)\} = d(x, y).$$

On the other hand, by the strong triangle inequality again, we have

$$d(x, y) \leq \max\{d(x, z), d(z, y)\} = \max\{d(x, z), d(y, z)\} \leq d(x, y).$$

This shows that $d(x, y) = \max\{d(x, z), d(y, z)\}$. Thus either $d(x, z) = d(x, y) \geq d(y, z)$ or $d(y, z) = d(x, y) \geq d(x, z)$. $\square$

Proposition 2.4. Let (X, d) be an ultra-metric space. Let D_i be (open or closed) ball in X for $i = 1, 2$. If $D_1 \cap D_2 \neq \emptyset$, then either $D_1 \subseteq D_2$ or $D_2 \subseteq D_1$.

Proof. Suppose that D_i has center x_i and radius r_i for $i = 1, 2$. Let $y \in D_1 \cap D_2$. We have

$$d(x_1, x_2) \leq \max\{d(x_1, y), d(y, x_2)\}.$$

Without loss of generality, we may assume that $d(x_1, x_2) \leq d(x_1, y)$. It follows that $x_2 \in D_1$ since $d(x_1, y) < r_1$ (or $\leq r_1$).

If there exists $z \in D_2 \setminus D_1$, we claim that $D_1 \subseteq D_2$. We have $d(x_1, z) > d(x_1, x_2)$. Then by [Proposition 2.3](#),

$$r_1 \leq d(x_1, z) = d(x_2, z) \leq r_2.$$

In particular, if D_2 is an open ball, then we have strict inequality $r_1 < r_2$. For any $w \in D_1$, we have

$$d(x_2, w) \leq \max\{d(x_2, x_1), d(x_1, w)\} \leq r_1 \leq r_2.$$

Thus $w \in D_2$ whatever D_2 is open or closed, and it shows that $D_1 \subseteq D_2$. $\square$

Proposition 2.5. Let (X, d) be an ultra-metric space. Then both $B(x, r)$ and $E(x, r)$ are closed and open subsets of X for any $x \in X$ and $r > 0$.

Proof. We show that the sphere $S(x, r) := \{y \in X \mid d(x, y) = r\}$ is open in X . Note that if $y \in S(x, r)$, then for any $r' < r$, we have $B(y, r') \cap E(x, r) \neq \emptyset$ and $x \in E(x, r) \setminus B(y, r')$. Thus by [Proposition 2.4](#), we have $B(y, r') \subseteq E(x, r)$. If $B(y, r') \cap B(x, r) \neq \emptyset$, then by [Proposition 2.4](#) again, we have $B(y, r') \subseteq B(x, r)$. However, $y \in B(y, r') \setminus B(x, r)$, a contradiction. Thus $B(y, r') \subseteq E(x, r) \setminus B(x, r) = S(x, r)$. It yields that $S(x, r) = \bigcup_{y \in S(x, r)} B(y, r/2)$ is open in X .

Since $E(x, r) = B(x, r) \cup S(x, r)$ and $B(x, r) = E(x, r) \setminus S(x, r)$, both $B(x, r)$ and $E(x, r)$ are open and closed in X . $\square$

Corollary 2.6. Let (X, d) be an ultra-metric space. Then X is totally disconnected, i.e., the only connected subsets of X are the set with at most one point.

Proof. Suppose that $S \subset X$ has at least two distinct points $x, y \in S$. Let $r := d(x, y) > 0$. Consider the open ball $B(x, r/2)$. By Proposition 2.5, $B(x, r/2)$ is both open and closed in X . Thus $B(x, r/2) \cap S$ is both open and closed in S , however, it is non-empty and not equal to S since it contains x but not y . This shows that S is disconnected. $\square$

Proposition 2.7. Let (X, d) be an ultra-metric space. A sequence $\{x_n\}$ in X is cauchy if and only if $d(x_n, x_{n+1}) \rightarrow 0$ as $n \rightarrow \infty$.

Proof. The necessity is true for all metric spaces. Suppose that $d(x_n, x) \rightarrow 0$ as $n \rightarrow \infty$. For any $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that $d(x_n, x_{n+1}) < \varepsilon$ for all $n \geq N$. For any $m, n \geq N$ with $m < n$, by the strong triangle inequality, we have

$$d(x_n, x_m) \leq \max\{d(x_n, x_{n-1}), d(x_{n-1}, x_m)\} \leq \max\{d(x_n, x_{n-1}), d(x_{n-1}, x_{n-2}), \dots, d(x_{m+1}, x_m)\} < \varepsilon.$$

This shows that $\{x_n\}$ is a cauchy sequence. $\square$

2.2 Algebra: ring of integers and residue field

Let $\mathbf{k}$ be a non-archimedean field. Then easily see that $\{x \in \mathbf{k} : \|x\| \leq 1\}$ is a subring of $\mathbf{k}$. Moreover, it is a local ring whose maximal ideal is $\{x \in \mathbf{k} : \|x\| < 1\}$.

Definition 2.8. Let $\mathbf{k}$ be a non-archimedean field. The *ring of integers* of $\mathbf{k}$ is defined as

$$\mathbf{k}^\circ := \{x \in \mathbf{k} : \|x\| \leq 1\}.$$

Its maximal ideal is

$$\mathbf{k}^{\circ\circ} := \{x \in \mathbf{k} : \|x\| < 1\}.$$

The *residue field* of $\mathbf{k}$ is defined as

$$\kappa_{\mathbf{k}} := \tilde{\mathbf{k}} := \mathbf{k}^\circ / \mathbf{k}^{\circ\circ}.$$

Set $I_{r,<} := B(0, r)$ and $I_{r,\leq} := E(0, r)$ for each $r \in (0, 1]$.

Proposition 2.9. The sets $I_{r,<}$ and $I_{r,\leq}$ are ideals of the ring of integers $\mathbf{k}^\circ$. Conversely, any ideal of $\mathbf{k}^\circ$ is of the form $I_{r,<}$ or $I_{r,\leq}$ for some $r \in (0, 1)$.

Proof. Let I be an ideal of $\mathbf{k}^\circ$. Set $r = \sup\{|a| : a \in I\}$ (resp. $r = \max\{|a| : a \in I\}$ when the maximum exists). Then, by definition, we have $I \subset I_{r,<}$ (resp. $I \subset I_{r,\leq}$). For every $x \in \mathbf{k}^\circ$ with $|x| < r$ (resp. $|x| \leq r$), there exists $a \in I$ such that $|x| \leq |a|$. Thus, $|x/a| \leq 1$ and so $x/a \in \mathbf{k}^\circ$. Since I is an ideal, we have $x = (x/a)a \in I$. Therefore, $I_{r,<} \subset I$ (resp. $I_{r,\leq} \subset I$). $\square$

Proposition 2.10. Let I_r be either $I_{r,<}$ or $I_{r,\leq}$ for each $r \in (0, 1)$. Suppose $\{r_n \in (0, 1)\}_{n \in \mathbb{N}}$ is a decreasing sequence converging to 0. Then the completion $\hat{\mathbf{k}}$ of $\mathbf{k}$ is isomorphic to the projective

limit

$$\widehat{\mathbf{k}}^\circ \cong \varprojlim_{n \in \mathbb{N}} \mathbf{k}^\circ / I_{r_n}.$$

Proof. For every $x \in \widehat{\mathbf{k}}^\circ$, there exists a cauchy sequence $\{x_m\}_{m \in \mathbb{N}}$ in $\mathbf{k}^\circ$ converging to x . Since $\{r_n\}_{n \in \mathbb{N}}$ converges to 0, for each $n \in \mathbb{N}$, there exists $M_n \in \mathbb{N}$ such that for all $m, m' \geq M_n$, we have $|x_m - x_{m'}| < r_n$. Thus, the sequence $\{x_m + I_{r_n}\}_{m \in \mathbb{N}}$ is eventually constant in $\mathbf{k}^\circ / I_{r_n}$. Define a map

$$\Phi : \widehat{\mathbf{k}}^\circ \rightarrow \varprojlim_{n \in \mathbb{N}} \mathbf{k}^\circ / I_{r_n}, \quad x \mapsto \left(\lim_{m \rightarrow \infty} x_m + I_{r_n} \right)_{n \in \mathbb{N}}.$$

It is straightforward to verify that Φ is a well-defined ring homomorphism.

Conversely, for every $(a_n + I_{r_n})_{n \in \mathbb{N}} \in \varprojlim_{n \in \mathbb{N}} \mathbf{k}^\circ / I_{r_n}$, we can choose a representative $a_n \in \mathbf{k}^\circ$ for each n . We claim that the sequence $\{a_n\}_{n \in \mathbb{N}}$ is a cauchy sequence in $\mathbf{k}^\circ$. Indeed, for every $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that $r_N < \varepsilon$. For all $m, n \geq N$, since $a_n + I_{r_n}$ maps to $a_m + I_{r_m}$ under the natural projection, we have $|a_n - a_m| < r_N < \varepsilon$. Thus, $\{a_n\}_{n \in \mathbb{N}}$ converges to some $x \in \widehat{\mathbf{k}}^\circ$. Easily see that the limit x is independent of the choice of representatives $\{a_n\}_{n \in \mathbb{N}}$. This gives a map

$$\Psi : \varprojlim_{n \in \mathbb{N}} \mathbf{k}^\circ / I_{r_n} \rightarrow \widehat{\mathbf{k}}^\circ, \quad (a_n + I_{r_n})_{n \in \mathbb{N}} \mapsto \lim_{n \rightarrow \infty} a_n.$$

Direct verification shows that $\Psi = \Phi^{-1}$. □

Corollary 2.11. Let $\mathbf{k}$ be a non-archimedean field and $\widehat{\mathbf{k}}$ its completion. Then the residue field $\widehat{\mathbf{k}}_{\mathbf{k}} \cong \mathbf{k}_{\mathbf{k}}$ under the natural embedding $\mathbf{k}^\circ \hookrightarrow \widehat{\mathbf{k}}^\circ$.

Corollary 2.12. Let $\mathbf{k}$ be a non-archimedean field and $\widehat{\mathbf{k}}$ its completion. Then the valuation group $|\widehat{\mathbf{k}}^\times|$ of $\widehat{\mathbf{k}}$ is equal to the valuation group $|\mathbf{k}^\times|$ of $\mathbf{k}$.

Proof. Note that

$$\begin{aligned} r \in |\widehat{\mathbf{k}}^\times| &\iff I_{r, <} \subsetneq I_{r, \leq} \text{ in } \widehat{\mathbf{k}}^\circ \\ &\iff \widehat{\mathbf{k}}^\circ / I_{r, <} \rightarrow \widehat{\mathbf{k}}^\circ / I_{r, \leq} \text{ is not an isomorphism} \\ &\iff \mathbf{k}^\circ / I_{r, <} \rightarrow \mathbf{k}^\circ / I_{r, \leq} \text{ is not an isomorphism} \\ &\iff I_{r, <} \subsetneq I_{r, \leq} \text{ in } \mathbf{k}^\circ \\ &\iff r \in |\mathbf{k}^\times|. \end{aligned}$$

□

Proposition 2.13. Let $\mathbf{k}$ be a non-archimedean field with non-trivial valuation. Then $\mathbf{k}^\circ$ is totally bounded iff $\mathbf{k}^\circ / I_{r, <}$ and $\mathbf{k}^\circ / I_{r, \leq}$ are finite for each $r \in [0, 1]$. Moreover, if $\mathbf{k}$ is complete, then it is locally compact iff $\mathbf{k}^\circ / I_r$ is finite for each $r \in (0, 1)$.

Slogan “Locally compact $\iff$ pro-finite.”

Proof. We just prove the case for $I_r = I_{r,<}$. The case for $I_r = I_{r,\leq}$ is similar.

Suppose that $\mathbf{k}^\circ/I_r$ is finite for each $r \in [0, 1]$. Then for every $\varepsilon > 0$, there exists $r \in (0, 1)$ such that $r < \varepsilon$ and $\mathbf{k}^\circ/I_r$ is finite. Let $\{a_1 + I_r, \dots, a_n + I_r\}$ be the complete set of representatives of $\mathbf{k}^\circ/I_r$. Then the balls $B(a_i, r)$ for $i = 1, \dots, n$ cover $\mathbf{k}^\circ$.

Conversely, suppose that $\mathbf{k}^\circ/I_r$ is infinite for some $r \in [0, 1]$. Then there exists an infinite set $\{a_n\}$ with $|a_n| \in [r, 1]$ such that their images in $\mathbf{k}^\circ/I_r$ are distinct. In particular, for every $m \neq n$, we have $|a_n - a_m| \geq r$. Any subsequence of $\{a_n\}$ is not cauchy. Thus, $\mathbf{k}^\circ$ is not totally bounded. $\square$

Proposition 2.14. The ring $\mathbf{k}^\circ$ is noetherian iff $\mathbf{k}$ is a discrete valuation field.

Proof. Note that $|\mathbf{k}^\times| \subset \mathbb{R}_{>0}$ is a multiplicative subgroup. If $\mathbf{k}$ is not a discrete valuation field, then $|\mathbf{k}^\times|$ is dense in $\mathbb{R}_{>0}$. In particular, there exists a strictly ascending sequence $r_n \in |\mathbf{k}^\times| \cap (0, 1)$. Then the ideals $I_{r_n, \leq}$ form a strictly ascending chain of ideals in $\mathbf{k}^\circ$.

The converse is standard since now $\mathbf{k}^\circ$ is a discrete valuation ring. $\square$

Proposition 2.15. Let $\mathbf{k}$ be a complete non-archimedean field. Then $\mathbf{k}$ is locally compact iff $\mathbf{k}$ is a discrete valuation field and its residue field $\mathcal{K}_{\mathbf{k}}$ is finite.

Proof. The necessity follows from Proposition 2.13. For the sufficiency, suppose that $\mathbf{k}$ is a discrete valuation field whose residue field $\mathcal{K}_{\mathbf{k}}$ is finite. Let $\pi \in \mathbf{k}^\circ$ be a uniformizer. We only need to show that $\mathbf{k}^\circ/\pi^n \mathbf{k}^\circ$ is finite for each $n \in \mathbb{N}$. Note that there is an isomorphism

$$\pi^{n-1} \mathbf{k}^\circ / \pi^n \mathbf{k}^\circ \cong \mathcal{K}_{\mathbf{k}}, \quad x + \pi^n \mathbf{k}^\circ \mapsto \overline{x/\pi^{n-1}}.$$

Thus, by induction on n , we conclude that $\mathbf{k}^\circ/\pi^n \mathbf{k}^\circ$ is finite. $\square$

2.3 Hensel's Lemma

Theorem 2.16 (Hensel's lemma). Let $\mathbf{k}$ be a complete non-archimedean field and $F(T) \in \mathbf{k}^\circ[T]$ a monic polynomial. Suppose that the reduction $f(T) \in \mathcal{K}_{\mathbf{k}}[T]$ of $F(T)$ factors as

$$f(T) = g(T)h(T),$$

where $g(T), h(T) \in \mathcal{K}_{\mathbf{k}}[T]$ are monic polynomials that are coprime in $\mathcal{K}_{\mathbf{k}}[T]$. Then there exist monic polynomials $G(T), H(T) \in \mathbf{k}^\circ[T]$ such that

$$F(T) = G(T)H(T),$$

and the reductions of $G(T), H(T)$ in $\mathcal{K}_{\mathbf{k}}[T]$ are $g(T), h(T)$ respectively.

Proof. Since $\gcd(g, h) = 1$ in $\mathcal{K}_{\mathbf{k}}[T]$, there exist polynomials $u(T), v(T) \in \mathcal{K}_{\mathbf{k}}[T]$ such that $ug + vh = 1$ and $\deg u < \deg h, \deg v < \deg g$. Choose lifts $G_0(T), H_0(T), U(T), V(T) \in \mathbf{k}^\circ[T]$ of $g(T), h(T), u(T), v(T)$ respectively preserving their degrees such that G_0 and H_0 are monic. Then there exist $r < 1$ such that

$$U(T)G_0(T) + V(T)H_0(T) \equiv 1 \pmod{I_r}, \quad F(T) - G_0(T)H_0(T) \equiv 0 \pmod{I_r},$$

where $I_r = \{a \in \mathbf{k}^\circ : |a| < r\}$.

We will construct a sequence of monic polynomials $\{G_n(T)\}_{n \in \mathbb{N}}$ and $\{H_n(T)\}_{n \in \mathbb{N}}$ in $\mathbf{k}^\circ[T]$ such that for each $n \in \mathbb{N}$,

$$G_n(T) \equiv G_{n-1}(T) \pmod{I_{r^n}}, \quad H_n(T) \equiv H_{n-1}(T) \pmod{I_{r^n}},$$

and

$$F(T) - G_n(T)H_n(T) \equiv 0 \pmod{I_{r^{n+1}}}.$$

If we have such sequences, then their coefficients converge in the complete ring $\mathbf{k}^\circ$. Let $G(T)$ and $H(T)$ be the limits of $\{G_n(T)\}$ and $\{H_n(T)\}$ respectively. Then we have $F(T) = G(T)H(T)$ and the reductions of $G(T), H(T)$ in $\mathcal{K}_{\mathbf{k}}[T]$ are $g(T), h(T)$ respectively.

The case $n = 0$ is done by the above construction. Now suppose that we have constructed $G_n(T)$ and $H_n(T)$ for some $n \geq 0$. Since $G_n - G_0 \equiv 0 \pmod{I_r}$ and $H_n - H_0 \equiv 0 \pmod{I_r}$, we have

$$UG_n + VH_n = UG_0 + VH_0 + U(G_n - G_0) + V(H_n - H_0) \equiv 1 \pmod{I_r}.$$

Set $\Delta_n(T) = F(T) - G_n(T)H_n(T) \in I_{r^{n+1}}[T]$ and $\epsilon_n = U\Delta_n, \delta_n = V\Delta_n \in I_{r^{n+1}}[T]$. Then we have

$$\begin{aligned} (G_n + \epsilon_n)(H_n + \delta_n) - F_n &= G_nH_n + G_n\delta_n + H_n\epsilon_n + \epsilon_n\delta_n - F_n \\ &= (UG_n + VH_n - 1)\Delta_n + \epsilon_n\delta_n \in I_{r^{n+2}}[T]. \end{aligned}$$

Thus, we can set

$$G_{n+1}(T) = G_n(T) + \epsilon_n(T), \quad H_{n+1}(T) = H_n(T) + \delta_n(T).$$

This finishes the induction. □

Corollary 2.17. Let $\mathbf{k}$ be a complete non-archimedean field and $F(T) \in \mathbf{k}^\circ[T]$ a monic polynomial. Suppose that the reduction $f(T) \in \mathcal{K}_{\mathbf{k}}[T]$ of $F(T)$ has a simple root $a \in \mathcal{K}_{\mathbf{k}}$. Then there exists a root $\alpha \in \mathbf{k}^\circ$ of $F(T)$ whose reduction is a .

Proof. Since a is a simple root of $f(T)$, we have the factorization $f(T) = (T - a)h(T)$ for some monic polynomial $h(T) \in \mathcal{K}_{\mathbf{k}}[T]$ with $h(a) \neq 0$. Then the result follows from [Theorem 2.16](#). □

2.4 Newton polygons

Yang: To be filled.

3 Finite field extensions

3.1 Finite-dimensional vector space

Definition 3.1. Let $\mathbf{k}$ be a valuation field and V a vector space over $\mathbf{k}$. A *norm* on V is a function $\|\cdot\| : V \rightarrow \mathbb{R}_{\geq 0}$ satisfying the following properties for all $x, y \in V$ and $a \in \mathbf{k}$:

- (a) $\|x\| = 0$ if and only if $x = 0$;
- (b) $\|ax\| = |a| \cdot \|x\|$;

(c) $\|x + y\| \leq \|x\| + \|y\|$.

Example 3.2. Let $\mathbf{k}$ be a valuation field and V a finite-dimensional vector space over $\mathbf{k}$ with basis $\{e_1, e_2, \dots, e_n\}$. For any $x = a_1e_1 + a_2e_2 + \dots + a_ne_n \in V$, define

$$\|x\|_{\max} := \max_{1 \leq i \leq n} |a_i|.$$

Then $\|\cdot\|_{\max}$ is a norm on V , called the *maximal norm* with respect to the basis $\{e_1, e_2, \dots, e_n\}$.

Example 3.3. Setting as in [Example 3.2](#), for any $x = a_1e_1 + a_2e_2 + \dots + a_ne_n \in V$, define

$$\|x\|_1 := |a_1| + |a_2| + \dots + |a_n|.$$

Then $\|\cdot\|_1$ is also a norm on V .

Definition 3.4. Let $\mathbf{k}$ be a valuation field and V a vector space over $\mathbf{k}$. Two norms $\|\cdot\|_1$ and $\|\cdot\|_2$ on V are said to be *equivalent* if there exist positive constants $C_1, C_2 > 0$ such that for all $x \in V$,

$$C_1\|x\|_1 \leq \|x\|_2 \leq C_2\|x\|_1.$$

Lemma 3.5. Let $\mathbf{k}$ be a valuation field and V a vector space over $\mathbf{k}$. Two norms $\|\cdot\|_1$ and $\|\cdot\|_2$ on V are equivalent if and only if they induce the same topology on V .

Proof. The sufficiency is clear. Now suppose that $\|\cdot\|_1$ and $\|\cdot\|_2$ induce the same topology on V . Hence the unit open ball with respect to $\|\cdot\|_1$ contains a unit open ball with respect to $\|\cdot\|_2$. That is,

$$\{x \in V : \|x\|_1 < 1\} \supseteq \{x \in V : \|x\|_2 < C\}.$$

Then for every $x \in V$ with $\|x\|_1 = 1$, we have $\|x\|_2 \geq C = C\|x\|_1$. By scaling, we get that for every $x \in V$,

$$\|x\|_2 \geq C\|x\|_1.$$

Similar for the other direction, we conclude that $\|\cdot\|_1$ and $\|\cdot\|_2$ are equivalent. $\square$

Proposition 3.6. Let V be a normed finite-dimensional vector space over a complete valuation field $\mathbf{k}$. Then V is complete.

Proof. Yang: To be added. $\square$

Theorem 3.7. Let V be a finite-dimensional vector space over a complete field $\mathbf{k}$. Then all norms on V are equivalent.

Proof. Fix a basis $\{e_1, e_2, \dots, e_n\}$ of V and let $\|\cdot\|_{\max}$ be the maximal norm with respect to this basis as in [Example 3.2](#). Let $\|\cdot\|$ be any norm on V . It suffices to show that $\|\cdot\|$ and $\|\cdot\|_{\max}$ are equivalent. First we have

$$\|y\| \leq \sum_{i=1}^n |a_i| \|e_i\| \leq \left(\sum_{i=1}^n \|e_i\| \right) \|y\|_{\max}$$

for any $y = a_1e_1 + a_2e_2 + \dots + a_ne_n \in V$. It remains to show that there exists a constant $C > 0$ such

that for any $y \in V$,

$$\|y\|_{\max} \leq C\|y\|.$$

Yang: To be added. □

Remark 3.8. If the base field $\mathbf{k}$ is not complete, then [Theorem 3.7](#) may fail. For example, let $\mathbf{k} = \mathbb{Q}$ with the usual absolute value, and let $V = \mathbb{Q}[\alpha]$ with $\alpha^2 - \alpha - 1 = 0$. There are two embeddings of V into $\mathbb{R}$:

$$\iota_1 : a + b\alpha \mapsto a + b\frac{1+\sqrt{5}}{2}, \quad \iota_2 : a + b\alpha \mapsto a + b\frac{1-\sqrt{5}}{2}.$$

Define two norms on V by

$$\|x\|_1 := |\iota_1(x)|, \quad \|x\|_2 := |\iota_2(x)|,$$

where $|\cdot|$ is the usual absolute value on $\mathbb{R}$. Then $\|\cdot\|_1$ and $\|\cdot\|_2$ are not equivalent since $\iota_2(\alpha^n) \rightarrow 0$ as $n \rightarrow \infty$ while $\iota_1(\alpha^n) \rightarrow \infty$.

The following lemma is a classical result in functional analysis, which will be used in the next subsection.

Lemma 3.9. Let $\mathbf{k}$ be a complete field and V a normed finite-dimensional vector space over $\mathbf{k}$. Then

$$\|\cdot\| : \text{End}_{\mathbf{k}}(V) \rightarrow \mathbb{R}_{\geq 0}, \quad T \mapsto \sup_{x \in V \setminus \{0\}} \frac{\|T(x)\|}{\|x\|}$$

defines a norm on the $\mathbf{k}$ -vector space $\text{End}_{\mathbf{k}}(V)$ satisfying

$$\|AB\| \leq \|A\| \cdot \|B\|, \quad \forall A, B \in \text{End}_{\mathbf{k}}(V).$$

Proof. First we show the existence of the supremum, i.e., there exists $C > 0$ such that for all $x \in V \setminus \{0\}$, $\|T(x)\| \leq C\|x\|$. Fix a basis $\{e_1, e_2, \dots, e_n\}$ of V and let $\|\cdot\|_{\max}$ be the maximal norm with respect to this basis. Since all norms on V are bounded by each other by [Theorem 3.7](#), we only need to show that there exists $C > 0$ such that for all $x \in V \setminus \{0\}$, $\|T(x)\|_1 \leq C\|x\|_{\max}$. Write $T(e_i) = \sum_{j=1}^n a_{ij}e_j$ for $1 \leq i \leq n$. For any $x = \sum_{i=1}^n x_i e_i \in V$, we have

$$\|T(x)\|_1 = \left\| \sum_{j=1}^n \left(\sum_{i=1}^n a_{ij}x_i \right) e_j \right\|_1 = \sum_{j=1}^n \left| \sum_{i=1}^n a_{ij}x_i \right| \leq \left(\sum_{1 \leq i, j \leq n} |a_{ij}| \right) \|x\|_{\max}.$$

Thus the supremum is finite.

The linearity and positive-definiteness of $\|\cdot\|$ are clear. It remains to show the triangle inequality and sub-multiplicativity. For any $A, B \in \text{End}_{\mathbf{k}}(V)$, we have

$$\frac{\|(A+B)(x)\|}{\|x\|} = \frac{\|A(x)\|}{\|x\|} + \frac{\|B(x)\|}{\|x\|} \leq \|A\| + \|B\|.$$

Taking supremum over all $x \in V \setminus \{0\}$ gives $\|A+B\| \leq \|A\| + \|B\|$. We have

$$\|AB(x)\| \leq \|A\| \cdot \|B(x)\| \leq \|A\| \cdot \|B\| \cdot \|x\|$$

and hence $\|AB(x)\|/\|x\| \leq \|A\| \cdot \|B\|$. Taking supremum we get $\|AB\| \leq \|A\| \cdot \|B\|$. □

3.2 Finite field extensions

Lemma 3.10. Let $\mathbf{k}$ be a complete field and $\mathbf{l}$ a finite extension of $\mathbf{k}$. Then there exists an absolute value on $\mathbf{l}$ extending the absolute value on $\mathbf{k}$.

Proof. Fix a norm $\|\cdot\|_V$ on the $\mathbf{k}$ -vector space $V = \mathbf{l}$. The norm $\|\cdot\|_V$ induces an operator norm $\|\cdot\|_{\text{op}}$ on the $\mathbf{k}$ -vector space $\text{End}_{\mathbf{k}}(V)$ as in Lemma 3.9. For any $a \in \mathbf{l}$, let $\mu_a \in \text{End}_{\mathbf{k}}(V)$ be the $\mathbf{k}$ -linear map defined by multiplication by a . Note that $a \mapsto \mu_a$ gives an embedding of $\mathbf{k}$ -algebras and if $a \in \mathbf{k}$, $\|\mu_a\|_{\text{op}} = \|a\|_{\mathbf{k}}$. Thus the restriction of $\|\cdot\|_{\text{op}}$ to $\mathbf{l}$ gives an norm on $\mathbf{l}$ extending that on $\mathbf{k}$. The normed ring $(\mathbf{l}, \|\cdot\|_{\text{op}})$ is a Banach ring since it is a finite-dimensional vector space over the complete field $\mathbf{k}$. By ??, there exists a multiplicative seminorm $\|\cdot\|_{\mathbf{l}}$ on $\mathbf{l}$ bounded by $\|\cdot\|_{\text{op}}$. In particular, $\|\cdot\|_{\mathbf{l}}$ is bounded by $\|\cdot\|_{\mathbf{k}}$ on $\mathbf{k}$. On a field, if one norm is bounded by another norm, then they must be equal (consider the inverse elements). Thus $\|\cdot\|_{\mathbf{l}}$ extends the absolute value on $\mathbf{k}$. $\square$

Theorem 3.11. Let $\mathbf{k}$ be a complete field and $\mathbf{l}$ a finite extension of $\mathbf{k}$. Then the absolute value on $\mathbf{l}$ which extends the absolute value on $\mathbf{k}$ is uniquely determined by the absolute value on $\mathbf{k}$. Furthermore, we have

$$\|\cdot\|_{\mathbf{l}} = \|N_{\mathbf{l}/\mathbf{k}}(\cdot)\|_{\mathbf{k}}^{1/n},$$

where $n = [\mathbf{l} : \mathbf{k}]$ and $N_{\mathbf{l}/\mathbf{k}}$ is the norm map from $\mathbf{l}$ to $\mathbf{k}$.

Proof. Let $\|\cdot\|_{\mathbf{l}}$ be arbitrary absolute value on $\mathbf{l}$ extending that on $\mathbf{k}$. We will show that $\|\cdot\|_{\mathbf{l}}$ must be equal to $\|N_{\mathbf{l}/\mathbf{k}}(\cdot)\|_{\mathbf{k}}^{1/n}$. For any $a \in \mathbf{l}$, set $b = a^n / N_{\mathbf{l}/\mathbf{k}}(a) \in \mathbf{l}$. Then $N_{\mathbf{l}/\mathbf{k}}(b) = 1$ and

$$\|b\|_{\mathbf{l}} = \frac{\|a\|_{\mathbf{l}}^n}{\|N_{\mathbf{l}/\mathbf{k}}(a)\|_{\mathbf{k}}}.$$

Thus it suffices to show that $\|b\|_{\mathbf{l}} = 1$ whenever $N_{\mathbf{l}/\mathbf{k}}(b) = 1$.

Note that the norm map $N_{\mathbf{l}/\mathbf{k}} : \mathbf{l} \rightarrow \mathbf{k}$ is the determinant of the $\mathbf{k}$ -linear map $\mu_b \in \text{End}_{\mathbf{k}}(V)$ defined by multiplication by b . Hence it is continuous on $\mathbf{l}$ (since it is a polynomial in the entries of the matrix representation). If $\|b\|_{\mathbf{l}} < 1$, then $\|b^m\|_{\mathbf{l}} \rightarrow 0$ as $m \rightarrow \infty$. Thus $N_{\mathbf{l}/\mathbf{k}}(b^m) = \det(\mu_{b^m}) \rightarrow 0$ as $m \rightarrow \infty$, contradicting the fact that $N_{\mathbf{l}/\mathbf{k}}(b^m) = 1$ for all m . Similarly, if $\|b\|_{\mathbf{l}} > 1$, then just consider b^{-1} . $\square$

Proposition 3.12. Let $\mathbf{k}$ be an algebraically closed valuation field. Then its completion $\hat{\mathbf{k}}$ is also algebraically closed.

Proof. Let $f \in \hat{\mathbf{k}}[X]$ be a non-constant polynomial. We will show that f has a root in $\hat{\mathbf{k}}$. Take a sequence of polynomials $\{f_n\}_{n \in \mathbb{N}}$ in $\mathbf{k}[X]$ converging to f coefficient-wisely and of the same degree d . Since $\mathbf{k}$ is algebraically closed, each f_n splits completely in $\mathbf{k}$ and hence in $\hat{\mathbf{k}}$. Write $f_n(X) = \prod_{i=1}^d (X - \alpha_{n,i})$ with $\alpha_{n,i} \in \hat{\mathbf{k}}$.

Let $\mathbf{l}$ be a finite extension of $\hat{\mathbf{k}}$ such that f has a root α in $\mathbf{l}$. For every $\varepsilon > 0$, if there are infinitely many n such that $\alpha_{n,i} \notin B(\alpha, \varepsilon)$ for all $1 \leq i \leq d$, then we have $|f_n(\alpha)| \geq \varepsilon^d$ for infinitely many n , contradicting the fact that $f_n(\alpha) \rightarrow f(\alpha) = 0$. Thus for every $\varepsilon > 0$, there exists $N > 0$ such that for all $n \geq N$, there exists $1 \leq i \leq d$ with $\alpha_{n,i} \in B(\alpha, \varepsilon)$. That is, we can find a sequence $\alpha_{n,i_n} \in \mathbf{k}$ converging to α . Since $\hat{\mathbf{k}}$ is complete, we have $\alpha \in \hat{\mathbf{k}}$. $\square$

Theorem 3.13 (Krasner's lemma). Let $\mathbf{k}$ be a complete non-archimedean field, and $\alpha, \beta \in \overline{\mathbf{k}}$. Denote by $\alpha_1, \alpha_2, \dots, \alpha_n$ the conjugates of α over $\mathbf{k}$ with $\alpha_1 = \alpha$. If

$$|\beta - \alpha| < |\alpha - \alpha_i|, \quad \forall i = 2, \dots, n,$$

then $\alpha \in \mathbf{k}(\beta)$.

Proof. Yang: To be added. □

3.3 Ramification and inertia

In this subsection, we study the extensions of absolute values on finite field extensions. Note that we do not assume the base field to be complete.

Definition 3.14. Let L/K be a finite field extension, and $v \in M_K$ an absolute value on K . We denote by $w|v$ if $w \in M_L$ is an absolute value on L extending v . For each $w|v$, we define the *ramification index* $e(w|v)$ and the *inertia degree* $f(w|v)$ by

$$e(w|v) := [|\hat{L}^\times|_w : |\hat{K}^\times|_v], \quad f(w|v) := \frac{[\hat{L} : \hat{K}]}{e(w|v)},$$

where $\hat{K}$ and $\hat{L}$ are the completions of K and L with respect to v and w , respectively.

Lemma 3.15. Suppose that v is non-archimedean and κ_v and ℓ_w are the residue fields of K and L with respect to v and w , respectively. Then we have

$$f(w|v) = [\ell_w : \kappa_v].$$

Remark 3.16. Yang: To be added.

Theorem 3.17. Let L/K be a finite field extension, and $v \in M_K$ an absolute value on K . Then we have

$$\sum_{w|v} e(w|v)f(w|v) = [L : K].$$

Let L/K be a finite field extension, and $v \in M_K$ an absolute value on K . We have

$$L \otimes_K K_v \cong \prod_{w|v} L_w,$$

where the product is taken over all absolute values $w \in M_L$ extending v .

Theorem 3.18. Let $\mathbf{k}$ be a number field. Then

$$M_{\mathbf{k}}^\infty = \{\text{embeddings } \sigma : \mathbf{k} \rightarrow \mathbb{C}\}$$

and

$$M_{\mathbf{k}}^f = \{\text{non-zero prime ideals } \mathfrak{p} \subseteq \mathcal{O}_{\mathbf{k}}\}.$$

Yang: To be revised.

Proposition 3.19 (Product formula). Let $\mathbf{k}$ be a number field. For each $x \in \mathbf{k}^\times$, we have

$$\prod_{v \in M_{\mathbf{k}}} |x|_v^{n_v} = 1,$$

where

$$n_v := \begin{cases} [\mathbf{k}_v : \mathbb{R}], & v \in M_{\mathbf{k}}^\infty; \\ 1, & v \in M_{\mathbf{k}}^0. \end{cases}$$

Yang: To be revised.

Remark 3.20. Let L/K be a finite field extension, and $v \in M_K$ an absolute value on K . Suppose that v is non-archimedean. Yang: To be rewritten.

4 Example: p -adic fields

4.1 p -adic fields

Construction 4.1. Let K be a number field and $\mathfrak{p}$ be a prime ideal of the ring of integers $\mathcal{O}_K$ of K . Considering the localization $(\mathcal{O}_K)_{\mathfrak{p}}$ of $\mathcal{O}_K$ at $\mathfrak{p}$, which is a discrete valuation ring, denote by $v_{\mathfrak{p}} : K^\times \rightarrow \mathbb{Z}$ the corresponding discrete valuation. The p -adic absolute value on K associated to $\mathfrak{p}$ is defined as

$$|x|_{\mathfrak{p}} := N(\mathfrak{p})^{-v_{\mathfrak{p}}(x)}, \quad \forall x \in K,$$

where $N(\mathfrak{p}) := \#(\mathcal{O}_K/\mathfrak{p})$ is the norm of $\mathfrak{p}$.

The completion of K with respect to the p -adic absolute value $|\cdot|_{\mathfrak{p}}$ is denoted by $K_{\mathfrak{p}}$, called the p -adic field.

We just focus on the case $K = \mathbb{Q}$ and $\mathfrak{p} = (p)$ for a prime number p in the following.

Example 4.2. Let p be a prime number. For every $r \in \mathbb{Q}$, we can write r as $r = p^n \frac{a}{b}$, where $n \in \mathbb{Z}$ and $a, b \in \mathbb{Z}$ are integers not divisible by p . The p -adic absolute value on $\mathbb{Q}$ is defined as

$$|r|_p := p^{-n}.$$

The p -adic field $\mathbb{Q}_p$ can be described concretely as follows:

$$\mathbb{Q}_p = \left\{ \sum_{i=n}^{+\infty} a_i p^i \mid n \in \mathbb{Z}, a_i \in \{0, 1, \dots, p-1\} \right\}.$$

For $x = \sum_{i=n}^{+\infty} a_i p^i \in \mathbb{Q}_p$ with $a_n \neq 0$, its p -adic absolute value is given by $|x|_p = p^{-n}$. The operations of addition and multiplication on $\mathbb{Q}_p$ are defined similarly as those on decimal expansions.

Easily see that $|\mathbb{Q}_p^\times|_p = p^{\mathbb{Z}}$ and $\kappa_{\mathbb{Q}_p} \cong \mathbb{F}_p$.

Unlike the field of real numbers $\mathbb{R}$, the p -adic field $\mathbb{Q}_p$ has many finite extensions.

Proposition 4.3. The field $\mathbb{Q}_p$ has algebraic extensions of arbitrarily large degree.

Proof. Since there are infinitely many irreducible monic polynomials over the finite field $\mathbb{F}_p$, consider any lift of such an irreducible monic polynomial to a monic polynomial with coefficients in $\mathbb{Z}_p$. If the lift is not irreducible over $\mathbb{Q}_p$, then the factorization of the lift gives a nontrivial factorization of its reduction modulo p since the factors can be chosen to be monic and have coefficients in $\mathbb{Z}_p$, which contradicts the irreducibility of the original polynomial over $\mathbb{F}_p$. Thus, the lift is irreducible over $\mathbb{Q}_p$.

On the other hand, note that $|\mathbb{Q}_p^\times|_p = p^{\mathbb{Z}}$. It follows that $f(T) = T^n - p$ is irreducible over $\mathbb{Q}_p$ for every integer $n \geq 1$. Otherwise, suppose we have a monic factorization $f(T) = g(T)h(T)$ with $g(T), h(T) \in \mathbb{Z}_p[T]$ and $\deg g, \deg h < n$. Then by considering the reduction modulo p , we have $g(0), h(0) \equiv 0 \pmod{p}$. It follows that $|f(0)|_p = |g(0)h(0)|_p \leq p^{-2}$, which contradicts $|f(0)|_p = |p|_p = p^{-1}$. $\square$

Let

$$R := \bigcup_{n=1}^{+\infty} \mathbb{Q}_p(\mu_{p^n})$$

and

$$K := \bigcup_{n=1}^{+\infty} \mathbb{Q}_p\left(\bigcup_{p \nmid m} \mu_m\right).$$

Yang: R is a totally ramified extension and K is the maximal unramified extension.

4.2 Completion

Proposition 4.4. The algebraic closure $\overline{\mathbb{Q}_p}$ of $\mathbb{Q}_p$ is not complete with respect to the extension of the p -adic absolute value $|\cdot|_p$.

Proof. Let $(a_n)_{n \geq 1}$ be a sequence in $\overline{\mathbb{Q}_p}$ such that

$$[\mathbb{Q}_p(\{a_n\}) : \mathbb{Q}_p] = +\infty.$$

Set $s_1 = 0$ and

$$s_n = s_{n-1} + p^{k_n} a_n$$

for a suitable $k_n \in \mathbb{N}$ such that

$$|p^{k_n} a_n|_p \leq \frac{|p^{k_{n-1}} a_{n-1}|_p}{2}, \quad |p^{k_n} a_n|_p < |s_{n-1} - \sigma(s_{n-1})|_p \quad \forall \sigma \in \text{Gal}(\overline{\mathbb{Q}_p}/\mathbb{Q}_p) \text{ with } \sigma(s_{n-1}) \neq s_{n-1}.$$

Then the sequence $(s_n)_{n \geq 1}$ is a Cauchy sequence. Let $s \in \widehat{\overline{\mathbb{Q}_p}}$ be the limit of $(s_n)_{n \geq 1}$. We have $|s - s_{n-1}|_p = |p^{k_n} a_n|_p < |s_{n-1} - \sigma(s_{n-1})|_p$ for all $n \geq 1$ and all $\sigma \in \text{Gal}(\overline{\mathbb{Q}_p}/\mathbb{Q}_p)$ with $\sigma(s_{n-1}) \neq s_{n-1}$. By Krasner's lemma ([Theorem 3.13](#)), it follows that $s_n \in \mathbb{Q}_p(s)$ for all $n \geq 1$. Thus, we have

$$[\mathbb{Q}_p(s) : \mathbb{Q}_p] \geq [\mathbb{Q}_p(\{a_n\}) : \mathbb{Q}_p] = +\infty,$$

which implies that $s \notin \overline{\mathbb{Q}_p}$. $\square$

Construction 4.5. Let p be a prime number. The field $\mathbb{C}_p$ of p -adic complex numbers is defined as the completion of the algebraic closure of $\mathbb{Q}_p$ with respect to the unique extension of the p -adic absolute value $|\cdot|_p$ on $\mathbb{Q}_p$.

The field $\mathbb{C}_p$ is algebraically closed and complete with respect to $|\cdot|_p$ by [Proposition 3.12](#). By [Corollaries 2.11](#) and [2.12](#), we have

$$|\mathbb{C}_p^\times|_p = |\overline{\mathbb{Q}_p}^\times|_p = p^{\mathbb{Q}}, \quad \kappa_{\mathbb{C}_p} \cong \kappa_{\overline{\mathbb{Q}_p}} \cong \overline{\mathbb{F}_p}.$$

Proposition 4.6. The field $\mathbb{C}_p$ of p -adic complex numbers is not spherically complete.

Proof. Choose a strictly descending sequence $r_n \in |\mathbb{C}_p^\times|$ with $r_n \rightarrow r > 0$ as $n \rightarrow +\infty$. We inductively construct a collection of closed ball. Begin from r_0 , choose any nonempty closed ball E of radius r_0 . On E , choose two disjoint nonempty closed balls of radius r_1 and denote them by E_0 and E_1 . Inductively, for any binary sequence $a_1 a_2 \cdots a_n \in \{0, 1\}^n$, suppose that we have constructed the ball $E_{a_1 \dots a_n}$ has radius r_n . On $E_{a_1 \dots a_n}$, choose two disjoint nonempty closed balls of radius r_{n+1} and denote them by $E_{a_1 \dots a_n 0}$ and $E_{a_1 \dots a_n 1}$. In this way, we obtain a collection of nonempty closed balls

$$\{E_{a_1 \dots a_n} \mid n \geq 0, a_1 \cdots a_n \in \{0, 1\}^n\}.$$

For $a = a_1 a_2 \cdots \in \{0, 1\}^{\mathbb{N}}$, set

$$E_a := \bigcap_{n=0}^{+\infty} E_{a_1 \dots a_n}.$$

Then each E_a is a closed ball of radius r . And for $a \neq b \in \{0, 1\}^{\mathbb{N}}$, we have $E_a \cap E_b = \emptyset$. Note that $\overline{\mathbb{Q}}$ is dense in $\mathbb{C}_p$. Since $\overline{\mathbb{Q}}$ is countable and $\{0, 1\}^{\mathbb{N}}$ is uncountable, there exists some $a \in \{0, 1\}^{\mathbb{N}}$ such that $E_a \cap \overline{\mathbb{Q}_p} = \emptyset$. In ultra-metric spaces, closed balls are also open. It follows that $E_a = \emptyset$. $\square$

Remark 4.7. For a concrete example, see [p-adic fields for beginners](#).

Construction 4.8. Let p be a prime number. **Yang:** We construct the *spherically complete p -adic field Ω_p* . **Yang:** To be completed.

Yang: does Ω_p has the same card as $\mathbb{R}$?

References

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